Calibration process for single crystalline samples measured with six circle diffractometer

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1. Introduction

Nowadays X-ray diffraction has become one of the standard characterization techniques in material science studies. Three dimensional reciprocal space maps (3D RSM), for example, are routinely employed in the studies of single crystalline epitaxial thin films. These 3D RSM are usually recorded by rocking the specimen with six-circle diffractometer. Often the film Bragg peak is plotted with respected to substrate diffraction peak, so that their relative positions reflect the strain relaxations in the epitaxial processes. To correctly understand the difference in their positions, 3D RSM should be plotted along the certain crystalline directions. However, without proper calibration, q space can only be calculated in the lab coordinates, which defined by the motions of six circle diffractometer. To plot 3D RSM in crystalline coordinates, the orientation of the substrate needs to be calibrated and this calibration results needs to be considered in the three-dimensional reciprocal space map calculations.

As pointed out by the previous literatures, crystal orientation of the single crystalline materials can be identified by two Bragg peak positions with known miller indexes. This orientation is presented as the orientation matrix U_r, which satisfies the following relations:

$$\begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix} = U_r B \begin{bmatrix} h \\ k \\ l \end{bmatrix}$$

U_r matrix corresponds to the transformation matrix, which transforms the unit cell Cartesian coordinates to the lab coordinate space. Similarly we can also find a U matrix U_e, which transforms the unit cell Cartesian coordinates to the crystal coordinates. Our goal is then to find out the differences between these two matrixes and taken it into account during the 3D

RSM calculation.

For this aim, first detector has to be properly calibrated. Then the expected U matrix U_e is defined. Calculations of the q vector in the lab coordinates will be introduced next, so that the U_r matrix can be measured. Finally three dimensional reciprocal space maps will be calculated via the matrix transformation, and the correction methods which transforms it to the crystal coordinates will be introduced.

2. Detector calibration

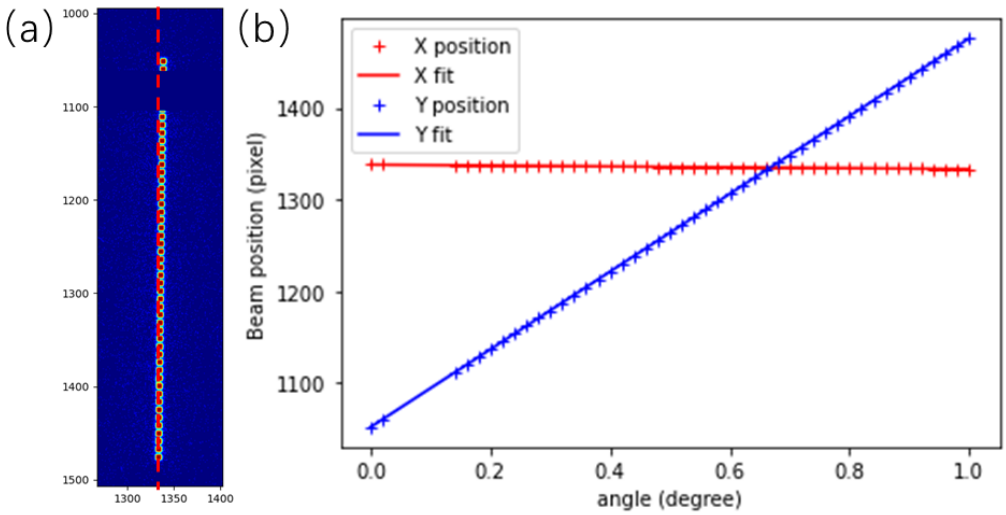


Figure 1: (a) Trace left be the direct beam positions on the detector during the calibration process, (b)

Before the sample calibration process, first detector must be calibrated. The calibration process is usually performed by scanning the delta motor of the detector arm. In this scan, detector moves with respect to the direct beam. In this way, the direct beam position changes on the detector at each scan position as shown in Fig. 1(a). The data treatment begins by estimating the weight center of the direct beam in each image. For the sample detector distance d, the inplane detector rotation det_rot, and the delta motor position delta, the weight center of the direction beam position X, Y can be written as

$$\mathrm{X} * pixelsize = \mathrm{d} * \sin(\det_rot) * \tan(delta) + X_0 * pixelsize$$

$$\mathrm{Y} * pixelsize = \mathrm{d} * \cos(\det_rot) * \tan(delta) + Y_0 * pixelsize$$

Thus, by linearly fitting X, Y position against $\tan(delta)$, we can get the detector distance d and the inplane detector rotation angle det_rot.

3. Calculating the B matrix

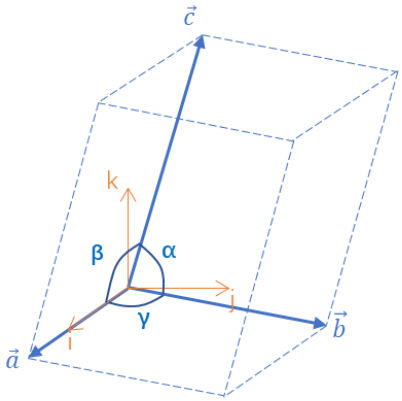


Figure 2: Unit cell defined in Cartesian coordinate frame

After the detector calibration, the first step of the sample calibration process is to calculate the B matrix. Unit cell, which are usually described with six parameters a, b, c, α, β, γ, must be expressed in Cartesian coordinates. As shown in the Fig. 2, here we define the aimed Cartesian coordinates according to the following rules:

- $\vec{a}, \vec{b}, \vec{c}$ form a right-handed set
- $\vec{a}$ is parallel to x axis
- $\vec{b}$ lies in xy plane, at an angle γ to $\vec{a}$
- $\vec{c}$ is pointed such that it has an angle β to $\vec{a}$, and an angle α to $\vec{b}$

This Cartesian coordinates is then called **unit cell Cartesian coordinates**. $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are unit vectors of this coordinates. According to this rule, the unit cell vector **a**, **b**, **c** can be written as:

$$\vec{a} = a\vec{\imath}$$
$$\vec{b} = b\cos\gamma\,\vec{\imath} + b\sin\gamma\,\vec{\jmath}$$
$$\vec{c} = x\vec{\imath} + y\vec{\jmath} + z\vec{k}$$

Here the c vector satisfies the following rules:

$$\vec{c}\cdot\vec{a} = xa = ca\cos\beta$$
$$\vec{c}\cdot\vec{b} = xb\cos\gamma + yb\sin\gamma = cb\cos\alpha$$
$$x^2 + y^2 + z^2 = c^2$$

Thus, the values of x, y, z can be written as

$$\mathrm{x} = \mathrm{c}\cos\beta$$
$$\mathrm{y} = \mathrm{c}\left(\frac{\cos\alpha - \cos\beta\cos\gamma}{\sin\gamma}\right)$$
$$\mathrm{z} = \mathrm{c}\left(\frac{1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma}{\sin^2\gamma}\right)$$

If we define a transformation matrix A which transforms the unit cell coordinates into this Cartesian coordinates. According to the equations before, this matrix can be written as

$$\mathrm{A} = [\vec{a}, \vec{b}, \vec{c}] = \begin{bmatrix} a_x & b_x & c_x \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{bmatrix}$$

$$= \begin{bmatrix} a & b\cos\gamma & c\cos\beta \\ 0 & b\sin\gamma & c\left(\dfrac{\cos\alpha - \cos\beta\cos\gamma}{\sin\gamma}\right) \\ 0 & 0 & c\left(\dfrac{1-\cos^2\alpha-\cos^2\beta-\cos^2\gamma+2\cos\alpha\cos\beta\cos\gamma}{\sin^2\gamma}\right) \end{bmatrix}$$

According to the definition of the reciprocal space vectors:

$$\overrightarrow{b_1}\cdot\vec{a}=2\pi;\ \overrightarrow{b_1}\cdot\vec{b}=0;\ \overrightarrow{b_1}\cdot\vec{c}=0;$$
$$\overrightarrow{b_2}\cdot\vec{a}=0;\ \overrightarrow{b_2}\cdot\vec{b}=2\pi;\ \overrightarrow{b_2}\cdot\vec{c}=0;$$
$$\overrightarrow{b_3}\cdot\vec{a}=0;\ \overrightarrow{b_3}\cdot\vec{b}=0;\ \overrightarrow{b_3}\cdot\vec{c}=2\pi;$$

Thus

$$\begin{bmatrix}\overrightarrow{b_1}\\ \overrightarrow{b_2}\\ \overrightarrow{b_3}\end{bmatrix}\begin{bmatrix}\vec{a} & \vec{b} & \vec{c}\end{bmatrix} = \begin{bmatrix}2\pi & 0 & 0\\ 0 & 2\pi & 0\\ 0 & 0 & 2\pi\end{bmatrix}$$

The B matrix, which consists of three reciprocal space vectors $\overrightarrow{b_1}, \overrightarrow{b_2}, \overrightarrow{b_3}$, can be calculated according to the following equations:

$$\mathrm{B} = \begin{bmatrix}\overrightarrow{b_1} & \overrightarrow{b_2} & \overrightarrow{b_3}\end{bmatrix} = 2\pi(A^{-1})^T$$

This B matrix transforms the miller indexes h, k, l to the q vector positions in the unit cell Cartesian coordinates.

4. Calculating the expected U matrix

During the experiment, the single crystalline specimen is glued on the sample stage of the six-circle diffractometer. In the ideal case, we would expect the surface direction of the specimen perpendicular to the surface of the sample stage, and the other crystalline direction along the X-ray beam. According to this ideal, we define a crystal coordinates, which in the ideal case is should be the same as our lab coordinates defined by the six-circle diffractometer. As illustrate in the Fig. 3, this **expected crystal coordinates** is a Cartesian coordinate, which obeys the following rules:

- The z axis is parallel to the surface direction of the sample;
- The x axis is an inplane crystal direction, which is along the beam.
- One of the in-plane directions should lay in the zx plane.

The U matrix transforms the **unit cell Cartesian coordinates** to this **expected crystal coordinates** can then be written as:

$$\begin{bmatrix}q_{ex}\\ q_{ey}\\ q_{ez}\end{bmatrix} = U_e B \begin{bmatrix}h\\ k\\ l\end{bmatrix}$$

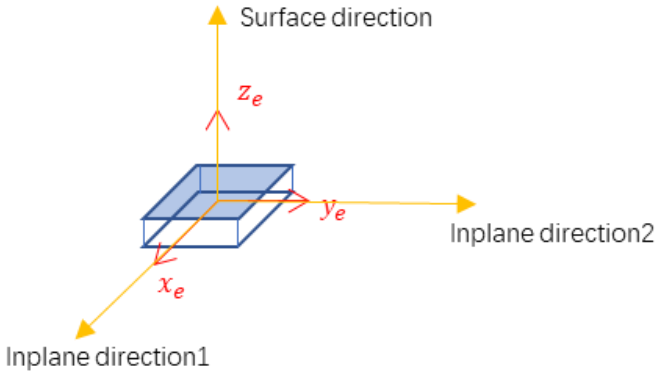


Figure 3: Schematic illustration of the expected crystal coordinates.

The surface direction $[h_s, k_s, l_s]$ can be expressed in unit cell Cartesian coordinates as

$$\overrightarrow{v_s} = B \begin{bmatrix} h_s \\ k_s \\ l_s \end{bmatrix}$$

In the expected crystal coordinates, this surface direction is parallel to the z axis.

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = U_e \frac{\overrightarrow{v_s}}{|\overrightarrow{v_s}|}$$

Similarly, the in-plane direction 1 can be expressed in unit cell Cartesian coordinates as

$$\overrightarrow{v_{i1}} = B \begin{bmatrix} h_1 \\ k_1 \\ l_1 \end{bmatrix}$$

Since the in-plane direction lays in the zx plane, the real in-plane components is

$$\overrightarrow{v_x} = \overrightarrow{v_{i1}} - \frac{\overrightarrow{v_{i1}} \cdot \overrightarrow{v_s}}{|\overrightarrow{v_s}|} \frac{\overrightarrow{v_s}}{|\overrightarrow{v_s}|}$$

According to the rules defined before, in the expected crystal coordinates, this direction is along the x axis.

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = U_e \frac{\overrightarrow{v_x}}{|\overrightarrow{v_x}|}$$

Finally the y direction is

$$\overrightarrow{v_y} = \frac{\overrightarrow{v_s}}{|\overrightarrow{v_s}|} \times \frac{\overrightarrow{v_{i1x}}}{|\overrightarrow{v_{i1x}}|}$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = U_e \frac{\overrightarrow{v_y}}{|\overrightarrow{v_y}|}$$

Thus, the expected transformation matrix U_e is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = U_e \begin{bmatrix} \frac{\overrightarrow{v_x}}{|\overrightarrow{v_x}|} & \frac{\overrightarrow{v_y}}{|\overrightarrow{v_y}|} & \frac{\overrightarrow{v_s}}{|\overrightarrow{v_s}|} \end{bmatrix}$$

$$U_e = \left(\begin{bmatrix} \frac{\overrightarrow{v_x}}{|\overrightarrow{v_x}|} & \frac{\overrightarrow{v_y}}{|\overrightarrow{v_y}|} & \frac{\overrightarrow{v_s}}{|\overrightarrow{v_s}|} \end{bmatrix} \right)^{-1}$$

5. Calculating the q vector

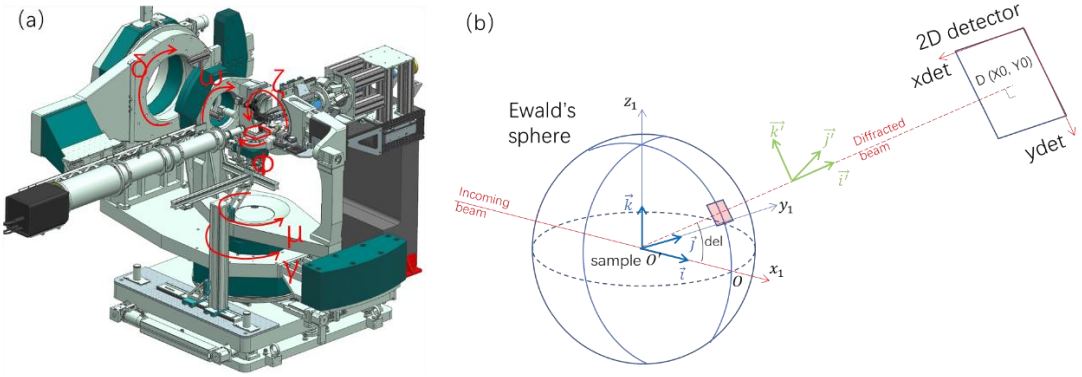


Figure 4: (a) Schematic overview of the six circle diffractometer at P10 beamline, (b) the direction of different coordinates.

As illustrated in Fig.4, the **real lab coordinates** is defined by the six circle diffractometer. Here the z direction is along the rotation axis of the γ motor, x direction is along the beam. According to this rule, we can transform the pixel position [X, Y] on the detector to the q value in this lab coordinates. The transformation can be summarized as the following steps:

1. The detector coordinates have to be projected on the surface of Ewald's sphere.

$$\overrightarrow{q_1} = [(Y_0 - Y) * pixelsize \quad (X_0 - X) * pixelsize \quad detector_distance] * \frac{1}{\rho} * \frac{2\pi}{\lambda}$$

where ρ is the distance between the detector pixel and the origin O'

$$\rho = \sqrt{detector_distance^2 + ((X_0 - X) * pixelsize)^2 + ((Y_0 - Y) * pixelsize)^2}$$

2. Then in-plane rotation of the detector is corrected. This is performed by a pure rotation coordinate transformation. As shown in the Fig. 5, this transformation of coordinates can be written as

$$x_2 = x_1$$
$$y_2 = y_1 \cos(\det_rot) + z_1 \sin(\det_rot)$$
$$z_2 = -y_1 \sin(\det_rot) + z_1 \cos(\det_rot)$$

Matrix form of this transformation can thus be expressed as:

$$\overrightarrow{q_2} = U_{dr}\overrightarrow{q_1} = \begin{bmatrix} \cos(\det_rot) & -\sin(\det_rot) & 0 \\ \sin(\det_rot) & \cos(\det_rot) & 0 \\ 0 & 0 & 1 \end{bmatrix} \overrightarrow{q_1}$$

3. The detector arm has two motors, delta and gamma. According to the stacking orders, the motor delta which is on top of gamma should be considered first. The corresponding coordinate transformation can be written as:

$$\overrightarrow{q_3} = U_{del}\overrightarrow{q_2} = \begin{bmatrix} \cos(delta) & 0 & \sin(delta) \\ 0 & 1 & 0 \\ -\sin(delta) & 0 & \cos(delta) \end{bmatrix} \overrightarrow{q_2}$$

4. The motor gamma has then to be considered:

$$\overrightarrow{q_4} = U_{gam}\overrightarrow{q_3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(gamma) & \sin(gamma) \\ 0 & -\sin(gamma) & \cos(gamma) \end{bmatrix} \overrightarrow{q_3}$$

5. The next step is to substrate the incoming beam vector, which corresponds the change of origin from **O'** to **O**. Then we get the diffraction vector q as:

$$\overrightarrow{q_5} = \overrightarrow{q_4} - \begin{bmatrix} 0 & 0 & \frac{2\pi}{\lambda} \end{bmatrix}$$

6. To transform the q vector to the coordinates defined by the surface of the sample stage. According to the stacking orders, the first motor that should be considered is mu.

$$\overrightarrow{q_6} = U_{mu}\overrightarrow{q_5} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(mu) & -sin(mu) \\ 0 & \sin(mu) & \cos(mu) \end{bmatrix} \overrightarrow{q_5}$$

7. The next motor for the coordinates transformation is omega.

$$\overrightarrow{q_7} = U_{om}\overrightarrow{q_6} = \begin{bmatrix} \cos(omega) & 0 & -\sin(omega) \\ 0 & 1 & 0 \\ \sin(omega) & 0 & \cos(omega) \end{bmatrix} \overrightarrow{q_6}$$

8. The chi motor is then considered. Here the zero position of the chi motor is considered to be at 90 degree. Then the aimed coordinates has similar orientations as our current one., and the chi rotation matrix can be written as:

$$\overrightarrow{q_8} = U_{chi}\overrightarrow{q_2} = \begin{bmatrix} \cos(chi) & -\sin(chi) & 0 \\ \sin(chi) & \cos(chi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \overrightarrow{q_7}$$

9. The phi motor, which is the top most sample motor, is finally considered:

$$\overrightarrow{q_r} = U_{phi}\overrightarrow{q_8} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(phi) & \sin(phi) \\ 0 & -\sin(phi) & \cos(phi) \end{bmatrix} \overrightarrow{q_8}$$

Here vector $\overrightarrow{q_r}$ is the q vector in our diffractometer coordinates (real lab coordinates), which can be used for the later experimental U matrix calculation.

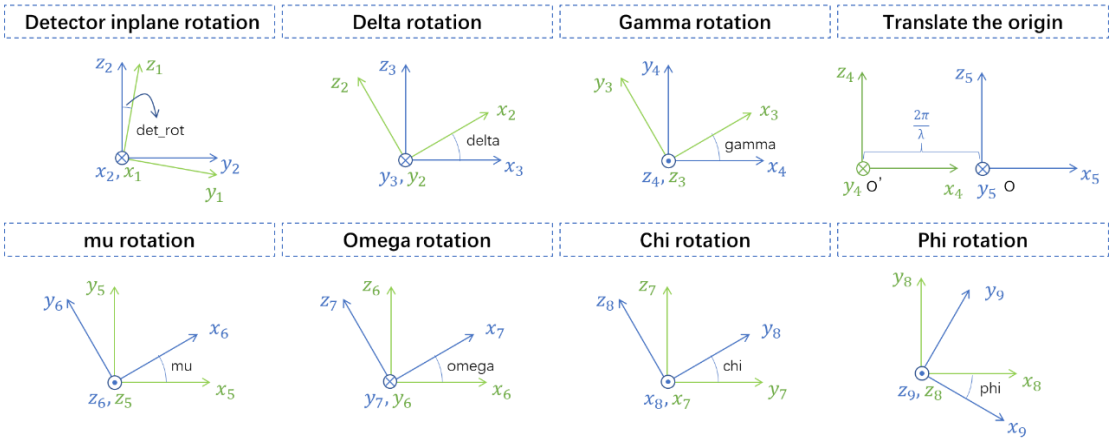


Figure 5: Illustrations of different rotations and their corresponding coordinate transformation.

6. Calculating the transformation matrix (Rocking curve with out of plane geometry)

Summarizing the equations in section 5, the vector $\mathbf{q}_r$ can be described according to the following equation:

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_{phi}U_{chi}U_{om}U_{mu}(U_{gam}U_{del}U_{dr}\overrightarrow{q_1} - \begin{bmatrix} 0 & 0 & \frac{2\pi}{\lambda} \end{bmatrix})$$

Assuming detector_distance >> (Y_0-Y)*pixelsize, (X_0-X)*pixelsize, the pixel distance $\rho \approx d$. With the unit defined as

$$\text{unit} = \frac{2\pi * pixelsize}{\lambda * detector_distance}$$

The previous equation can be written as

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_{phi}U_{chi}U_{om}U_{mu}(U_{gam}U_{del}U_{dr}\begin{bmatrix} (Y_0 - Y) \\ (X_0 - X) \\ \frac{detector_distance}{\text{pixelsize}} \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ \frac{detector_distance}{\text{pixelsize}} \end{bmatrix})$$

Here the omega angles of each detector image are expressed as om_0+Δom. om_0 is the beginning position of the omega motor, while Δom is the offset value of the actual omega. Then the U_{om} matrix can be separated as

$$U_{om} = \begin{bmatrix} \cos(om_0) & 0 & -\sin(om_0) \\ 0 & 1 & 0 \\ \sin(om_0) & 0 & \cos(om_0) \end{bmatrix}\begin{bmatrix} \cos(\Delta om) & 0 & -\sin(\Delta om) \\ 0 & 1 & 0 \\ \sin(\Delta om) & 0 & \cos(\Delta om) \end{bmatrix}$$

$$= U_{om0}\begin{bmatrix} \cos(\Delta om) & 0 & -\sin(\Delta om) \\ 0 & 1 & 0 \\ \sin(\Delta om) & 0 & \cos(\Delta om) \end{bmatrix}$$

This define two matrixes U_1, U_2 with the following equation:

$$U_1 = U_{mu}U_{gam}U_{del}U_{dr}$$
$$U_2 = U_{phi}U_{chi}U_{om0}$$

Thus, U_1 corresponds to combined rotation matrix of detector inplane rotation, delta motor, gamma motor and mu motor, while U_2 corresponds to the combined rotation matrix of omega motor, chi motor and phi motor.

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_2\begin{bmatrix} \cos(\Delta om) & 0 & -\sin(\Delta om) \\ 0 & 1 & 0 \\ \sin(\Delta om) & 0 & \cos(\Delta om) \end{bmatrix}(U_1\begin{bmatrix} (Y_0 - Y) \\ (X_0 - X) \\ \frac{detector_distance}{\text{pixelsize}} \end{bmatrix}$$

$$- \begin{bmatrix} 0 \\ -\sin(mu)\frac{detector_distance}{\text{pixelsize}} \\ \cos(mu)\frac{detector_distance}{\text{pixelsize}} \end{bmatrix})$$

Considering the U_1 matrix as

$$U_1 = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$$

Neglecting the constrains and the second order small value such as Δom*(X₀-X). The previous equation can be written as

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_2 \begin{bmatrix} \frac{\Delta om * detector_distance}{pixelsize}(\cos(mu) - a_9) + a_1(Y_0 - Y) + a_2(X_0 - X) \\ a_4(Y_0 - Y) + a_5(X_0 - X) \\ a_3\frac{\Delta om * detector_distance}{pixelsize} + a_7(Y_0 - Y) + a_8(X_0 - X) \end{bmatrix}$$

Here $\Delta \mathrm{om} = \mathrm{Z} * om_{step}$, where Z is the frame number in the omega scan and om_{step} is the step size of the scan:

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_2 \begin{bmatrix} \frac{om_{step} * detector_distance}{pixelsize}(\cos(mu) - a_9) & a_1 & a_2 \\ 0 & a_4 & a_5 \\ \frac{om_{step} * detector_distance}{pixelsize}a_3 & a_7 & a_8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} Z \\ Y \\ X \end{bmatrix}$$

Thus, a transformation matrix can be defined as

$$U_{transform} = U_2 \begin{bmatrix} \frac{om_{step} * detector_distance}{pixelsize}(\cos(mu) - a_9) & a_1 & a_2 \\ 0 & a_4 & a_5 \\ \frac{om_{step} * detector_distance}{pixelsize}a_3 & a_7 & a_8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

which transforms the indexes of the stacked detector images to the q values in the reciprocal space. Using the inverse matrix of this matrix, we can calculate the corresponding indexes of the q vector position, which is needed for the interpolation process. In this way, the 3D reciprocal space map can be plotted.

7. Calculating the transformation matrix (Rocking curve with in plane geometry)

Similarly for the in plane diffraction geometry, phi motor is often rotated. Phi angles of each detector image are expressed as phi₀+Δphi. The diffraction vector q in the

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\Delta phi) & \sin(\Delta phi) \\ 0 & -\sin(\Delta phi) & \cos(\Delta phi) \end{bmatrix} U_{phi0}U_{chi}U_{om}U_{mu}(U_{gam}U_{del}U_{dr} \begin{bmatrix} (Y_0 - Y) \\ (X_0 - X) \\ \frac{detector_distance}{\mathrm{pixelsize}} \end{bmatrix}$$

$$- \begin{bmatrix} 0 \\ 0 \\ \frac{detector_distance}{\mathrm{pixelsize}} \end{bmatrix})$$

Here we also define two matrixes U₁, U₂.

$$U_1 = U_{phi0}U_{chi}U_{om}U_{mu}U_{gam}U_{del}U_{dr}$$

$$U_2 = U_{phi0}U_{chi}U_{om}U_{mu}$$

The elementwise these two matrixes can be written as

$$U_1 = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$$

$$U_2 = \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix}$$

Then,

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\Delta phi) & \sin(\Delta phi) \\ 0 & -\sin(\Delta phi) & \cos(\Delta phi) \end{bmatrix} (U_1 \begin{bmatrix} (Y_0 - Y) \\ (X_0 - X) \\ \dfrac{detector_distance}{\text{pixelsize}} \end{bmatrix} - U_2 \begin{bmatrix} 0 \\ 0 \\ \dfrac{detector_distance}{\text{pixelsize}} \end{bmatrix})$$

Neglecting the constrains and the second order small value such as Δphi*(X_0-X). The previous equation can be written as

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = \begin{bmatrix} a_1(Y_0 - Y) + a_2(X_0 - X) \\ \dfrac{\Delta phi * detector_distance}{pixelsize}(a_9 - b_9) + a_4(Y_0 - Y) + a_5(X_0 - X) \\ \dfrac{\Delta phi * detector_distance}{pixelsize}(b_6 - a_6) + a_7(Y_0 - Y) + a_8(X_0 - X) \end{bmatrix}$$

Here $\Delta\text{phi} = \text{Z} * phi_{step}$, where Z is the frame number in the phi scan and phi_{step} is the step size of the scan:

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = \begin{bmatrix} 0 & a_1 & a_2 \\ \dfrac{phi_{step} * detector_distance}{pixelsize}(a_9 - b_9) & a_4 & a_5 \\ \dfrac{phi_{step} * detector_distance}{pixelsize}(b_6 - a_6) & a_7 & a_8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} Z \\ Y \\ X \end{bmatrix}$$

Thus, the transformation matrix can be defined as

$$U_{transform} = \begin{bmatrix} 0 & a_1 & a_2 \\ \dfrac{phi_{step} * detector_distance}{pixelsize}(a_9 - b_9) & a_4 & a_5 \\ \dfrac{phi_{step} * detector_distance}{pixelsize}(b_6 - a_6) & a_7 & a_8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

8. Calculating the transformation matrix (theta-2theta scan with out of plane geometry)

Here we again start from the following equation

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_{phi}U_{chi}U_{om}U_{mu}(U_{gam}U_{del}U_{dr} \begin{bmatrix} (Y_0 - Y) \\ (X_0 - X) \\ \dfrac{detector_distance}{\text{pixelsize}} \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ \dfrac{detector_distance}{\text{pixelsize}} \end{bmatrix})$$

In the theta-2theta scan, the delta angle varies twice as much as the omega angle. Here, the

omega angles for each detector image are expressed as $om_0+\Delta om$, and the delta angles as $del_0+2\Delta om$.Then the U_{om} matrix can be separated as

$$U_{om} = U_{om0}\begin{bmatrix}\cos(\Delta om) & 0 & -\sin(\Delta om)\\ 0 & 1 & 0\\ \sin(\Delta om) & 0 & \cos(\Delta om)\end{bmatrix}$$

The U_{del} matrix can be separated as

$$U_{del} = \begin{bmatrix}\cos(2\Delta om) & 0 & \sin(2\Delta om)\\ 0 & 1 & 0\\ -\sin(2\Delta om) & 0 & \cos(2\Delta om)\end{bmatrix}U_{del0}$$

For the simplicity here, the anlge mu and gamma here are assumed to be zeros. And we define two matrixes U_1, U_2 with the following equation:

$$U_1 = U_{del0}U_{dr}$$
$$U_2 = U_{phi}U_{chi}U_{om0}$$

The elementwise U_1 matrixes can be written as

$$U_1 = \begin{bmatrix}a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\\ a_7 & a_8 & a_9\end{bmatrix}$$

Neglecting the constrains and the second order small value such as $\Delta om*(X_0-X)$. The previous equation can be written as

$$\begin{bmatrix}q_{rz}\\ q_{ry}\\ q_{rx}\end{bmatrix} = U_2\begin{bmatrix}\frac{\Delta om * detector_distance}{pixelsize}(1+a_9) + a_1(Y_0-Y) + a_2(X_0-X)\\ a_4(Y_0-Y) + a_5(X_0-X)\\ -a_3\frac{\Delta om * detector_distance}{pixelsize} + a_7(Y_0-Y) + a_8(X_0-X)\end{bmatrix}$$

Here $\Delta\mathrm{om} = \mathrm{Z} * om_{step}$, where Z is the frame number in the omega scan and om_{step} is the step size of the scan:

$$\begin{bmatrix}q_{rz}\\ q_{ry}\\ q_{rx}\end{bmatrix} = U_2\begin{bmatrix}\frac{om_{step} * detector_distance}{pixelsize}(1+a_9) & a_1 & a_2\\ 0 & a_4 & a_5\\ -\frac{om_{step} * detector_distance}{pixelsize}a_3 & a_7 & a_8\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -1\end{bmatrix}\begin{bmatrix}Z\\ Y\\ X\end{bmatrix}$$

Thus, a transformation matrix can be defined as

$$U_{transform} = U_2\begin{bmatrix}\frac{om_{step} * detector_distance}{pixelsize}(1+a_9) & a_1 & a_2\\ 0 & a_4 & a_5\\ -\frac{om_{step} * detector_distance}{pixelsize}a_3 & a_7 & a_8\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -1\end{bmatrix}$$

Now consider a simpler case where the phi, chi, and detector rotation angles are all zero. In this case, the U1 matrix becomes the delta matrix, and the U2 matrix becomes the omega matrix. Thus, the above equation can be written as:

$$U_{transform}$$

$$= \begin{bmatrix} \cos(om_0) & 0 & -\sin(om_0) \\ 0 & 1 & 0 \\ \sin(om_0) & 0 & \cos(om_0) \end{bmatrix} \begin{bmatrix} \frac{om_{step} * detector_distance}{pixelsize}(1+\cos(del_0)) & \cos(del_0) & 0 \\ 0 & 0 & 1 \\ -\frac{om_{step} * detector_{distance}}{pixelsize}\sin(del_0) & -\sin(del_0) & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{om_{step} * detector_distance}{pixelsize}(\cos(om_0)+\cos(del_0-om_0)) & \cos(del_0-om_0) & 0 \\ 0 & 0 & 1 \\ \frac{om_{step} * detector_{distance}}{pixelsize}(\sin(om_0)-\sin(del_0-om_0)) & -\sin(del_0-om_0) & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

For the symmetric reflection, delta equal two times omega, $(\sin(om_0)-\sin(del_0-om_0))$ becomes zero, and the stacking direction is along the q_z direction, which corresponds to the simplest case in the theta-2theta scan.

Comparing to the pixel size, the distance in the stacking direction is

$$\text{rebinfactor} = 2(1+\cos(\text{delta}))\frac{om_{step} * detector_distance}{pixelsize}$$

Similar to the case of rocking curve, this parameter should be used to judge whether the step size in the scanning direction is fine enough.

9. Calculating the transformation matrix (theta-2theta scan with in plane geometry)

Similarly for the in plane diffraction geometry, phi motor and gamma is motor can be rotated. Due to the rotation direction, phi angles of each detector image are expressed as $phi_0+\Delta phi$, while the gamma angle is $gamma_0-2\Delta phi$. The diffraction vector q in the

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix}$$

$$= U_{phi0}\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\Delta phi) & \sin(\Delta phi) \\ 0 & -\sin(\Delta phi) & \cos(\Delta phi) \end{bmatrix} U_{chi}U_{om}U_{mu}\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(2\Delta phi) & -\sin(2\Delta phi) \\ 0 & \sin(2\Delta phi) & \cos(2\Delta phi) \end{bmatrix} U_{gam0}U_{del}U_{dr}\right.$$

$$\left.\begin{bmatrix} (Y_0-Y) \\ (X_0-X) \\ \frac{detector_distance}{\text{pixelsize}} \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ \frac{detector_distance}{\text{pixelsize}} \end{bmatrix}\right)$$

For the simplicity here, the anlge chi and omega here are assumed to be zeros. And we define two matrixes U_1, U_2 with the following equation:

$$U_1 = U_{gam0}U_{del}U_{dr}$$
$$U_2 = U_{phi0}U_{mu}$$

The elementwise U_1 matrixes can be written as

$$U_1 = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$$

Neglecting the constrains and the second order small value such as Δphi*(X_0-X). The previous equation can be written as

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_2 \begin{bmatrix} a_1(Y_0 - Y) + a_2(X_0 - X) \\ -\dfrac{\Delta phi * detector_distance}{pixelsize}(1 + a_9) + a_4(Y_0 - Y) + a_5(X_0 - X) \\ a_6 \dfrac{\Delta phi * detector_distance}{pixelsize} + a_7(Y_0 - Y) + a_8(X_0 - X) \end{bmatrix}$$

Here $\Delta\text{phi} = \text{Z} * phi_{step}$, where Z is the frame number in the omega scan and phi_{step} is the step size of the scan:

$$\begin{bmatrix} q_{rz} \\ q_{ry} \\ q_{rx} \end{bmatrix} = U_2 \begin{bmatrix} 0 & a_1 & a_2 \\ -\dfrac{phi_{step} * detector_distance}{pixelsize}(1 + a_9) & a_4 & a_5 \\ \dfrac{phi_{step} * detector_distance}{pixelsize} a_6 & a_7 & a_8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} Z \\ Y \\ X \end{bmatrix}$$

Thus, a transformation matrix can be defined as

$$U_{transform} = U_2 \begin{bmatrix} 0 & a_1 & a_2 \\ -\dfrac{phi_{step} * detector_distance}{pixelsize}(1 + a_9) & a_4 & a_5 \\ \dfrac{phi_{step} * detector_distance}{pixelsize} a_6 & a_7 & a_8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

10. Calculating misorientation matrix

If q vectors of two diffraction peaks are identified and their corresponding miller indexes are known, a U matrix can be calculated, which represents the orientation of the single crystalline substrate. This U matrix transforms the **unit cell Cartesian coordinates** to the **real lab coordinates** defined by the six circle diffractometer. Thus, this U matrix satisfies the following equation

$$\begin{bmatrix} q_{rx} \\ q_{ry} \\ q_{rz} \end{bmatrix} = U_r B \begin{bmatrix} h \\ k \\ l \end{bmatrix}$$

According to this equation, the U_r matrix can be extracted with least square fitting process. Comparing to the expected orientation matrix U_e, we define an misorientation matrix which transforms the real lab coordinates to the expected crystal coordinates. As the result the previous equations can be written as:

$$\begin{bmatrix} q_{ex} \\ q_{ey} \\ q_{ez} \end{bmatrix} = U_a \begin{bmatrix} q_{rx} \\ q_{ry} \\ q_{rz} \end{bmatrix} = U_e B \begin{bmatrix} h \\ k \\ l \end{bmatrix} = U_a U_r B \begin{bmatrix} h \\ k \\ l \end{bmatrix}$$

Thus,

$$U_a = U_e U_r^{-1}$$

Here we should note that in our previous calculation of the q vector in section 5. The q vector has an order of [q_z, q_y, q_x]. This choice facilities the later programming work in python. However, the U matrix is calculated to generate the q vector in [q_x, q_y, q_z] order. Thus, the final misorientation matrix U_a matrix has to be flipped to generate the real transformation matrix U_{af} for the python coding applications.

Then the real transformation matrix, which transforms the indexes of stacked detector images to the q vector should be written as

$$U_{real} = U_{af} U_{transform}$$

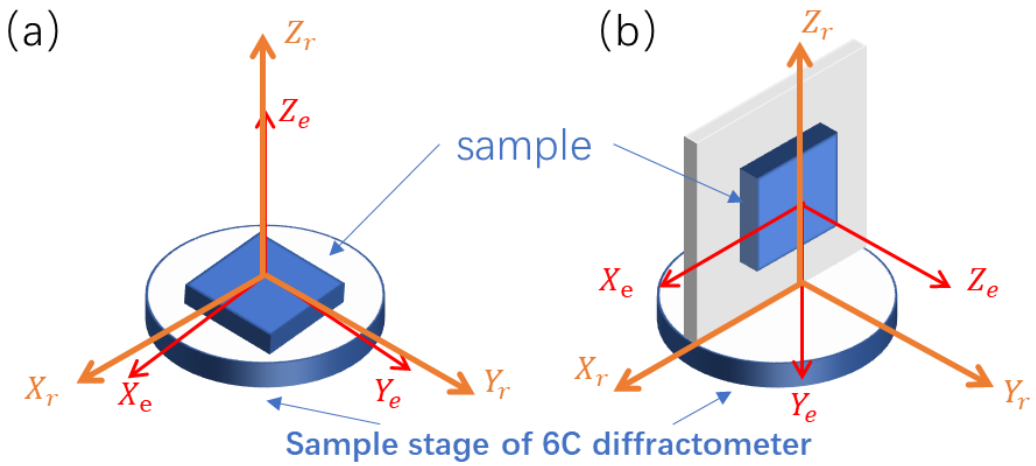


Figure 6: (a) Schematic illustration of the usual sample gluing scheme. (b) Schematic illustration of the vertical sample gluing scheme.

As schematically illustrated in Fig. 6(a), the orientation difference between the real lab coordinates and expected crystal coordinates are usually caused by the imperfect gluing process. The U_a matrix compensates this orientation difference and transforms the real lab coordinates to the crystal coordinates. Usually this orientation differences are small and the U_a matrix is close to the identity matrix. However, this does not need to be always the case. As shown in Fig. 6(b), in some special case, the sample is not glued directly on the sample stage, instead it is glued on some vertical stage to diffract in the in-plane geometry. Using the q vectors calculated in the real lab coordinates frame, the surface direction would not be along the z direction, which is different from our usual scheme. However, in this case, the U_a matrix would be close to

$$U_{af} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

With this transformation, we can be in the crystal coordinates, where the surface direction is along the q_z direction.